



POULTRY INTELLIGENCE & EARLY-WARNING SaaS

Week 2 — Define & Design

Designing the right solution for the validated poultry health problem.

PRD · User Journey · Wireframes · AI Workflow · Architecture

From one-time detector to an ongoing early-warning platform

This Week 2 document focuses on a vertical SaaS for the poultry industry. The core problem remains late detection of flock health issues — but the product is now designed as a continuous AI-powered poultry intelligence platform, not a one-time disease detector.



CORE PROBLEM

Small and medium-scale poultry farmers often detect disease and flock-health problems too late — leading to bird deaths, production drops, higher treatment costs, and significant financial losses.



PROPOSED SOLUTION

An affordable AI-powered poultry early-warning SaaS that monitors flock health and production data, detects unusual patterns, alerts farmers early, and recommends practical next steps.

Deliverables & milestone



PRD

Product Requirements
Document



User Journey

Stage-by-stage farmer flow



Wireframes

Phone-style screen layouts



AI Workflow

Detection-to-action pipeline



Architecture

System design diagram



MILESTONE

Demonstrate that AI is the right solution for this poultry use case.

Problem statement & product goal



1.1 PROBLEM STATEMENT

Small and medium-scale poultry farmers often detect disease and other flock-health problems too late. By the time visible symptoms appear, the issue may already have spread — causing mortality, lower egg or meat output, higher treatment costs, and significant financial losses.



1.2 PRODUCT GOAL

Help poultry farmers identify abnormal flock conditions earlier, take faster action, reduce preventable losses, and improve flock productivity and farm profitability.

1.3 SUCCESS METRICS

Time from abnormal flock signal to farmer action

Reduction in preventable mortality among active users

Weekly use of flock monitoring & check-in features

High-risk alerts confirmed useful by farmers/vets

Who we build for first



PRIMARY TARGET USER

Existing small & medium-scale poultry farmers who actively manage birds for eggs or meat and need an affordable way to monitor flock health, reduce mortality, and improve productivity.

SECONDARY USERS

New farmers needing guidance on health, vaccination, feeding.
Farm managers needing visibility across multiple flocks.
Veterinarians who receive escalated high-risk cases.

2.3 MVP SCOPE

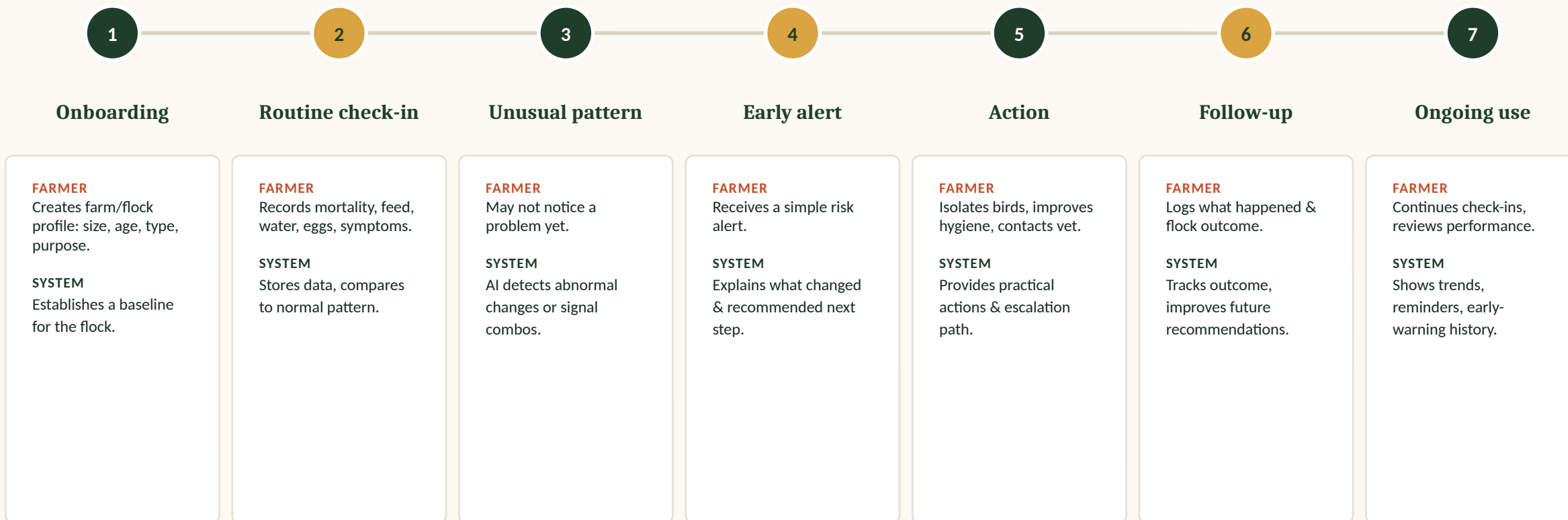
- Farm & flock profile: bird type, size, age, purpose, location
- AI anomaly detection across flock data
- Optional photo/symptom reporting
- Escalation to a vet for high-risk cases
- Daily check-in: mortality, feed, water, eggs, symptoms
- Plain-language risk alerts with next actions
- Case history & flock trend dashboard
- Offline-friendly capture with sync

Two farmer profiles, one platform

	 Existing Poultry Farmer	 New Poultry Farmer
Profile	Already manages an active flock for eggs or meat.	Starting or recently started poultry production.
Main goal	Protect flock, reduce losses, improve production & profit.	Avoid beginner mistakes and build healthy routines.
Pain points	Disease outbreaks, mortality, feed cost, limited vet access.	Uncertainty about feeding, vaccination and symptoms.
SaaS value	Earlier warning, trend visibility, faster action, records.	Guided monitoring, reminders, and practical next steps.

Primary MVP focus: design first for farmers already in the poultry business — they experience the problem today and can provide measurable evidence.

From onboarding to ongoing early warning



Editable phone-style screens

Three core mobile screens covering the daily flow: check-in entry, flock dashboard, and an early-warning alert.

Daily Check-in

Mortality count

Feed & water level

Egg / growth output

Symptom notes

Photo (optional)

Flock Dashboard

Flock health trend

Risk status: Watch

Recent check-ins

Reminders

Early Alert

⚠ High risk detected

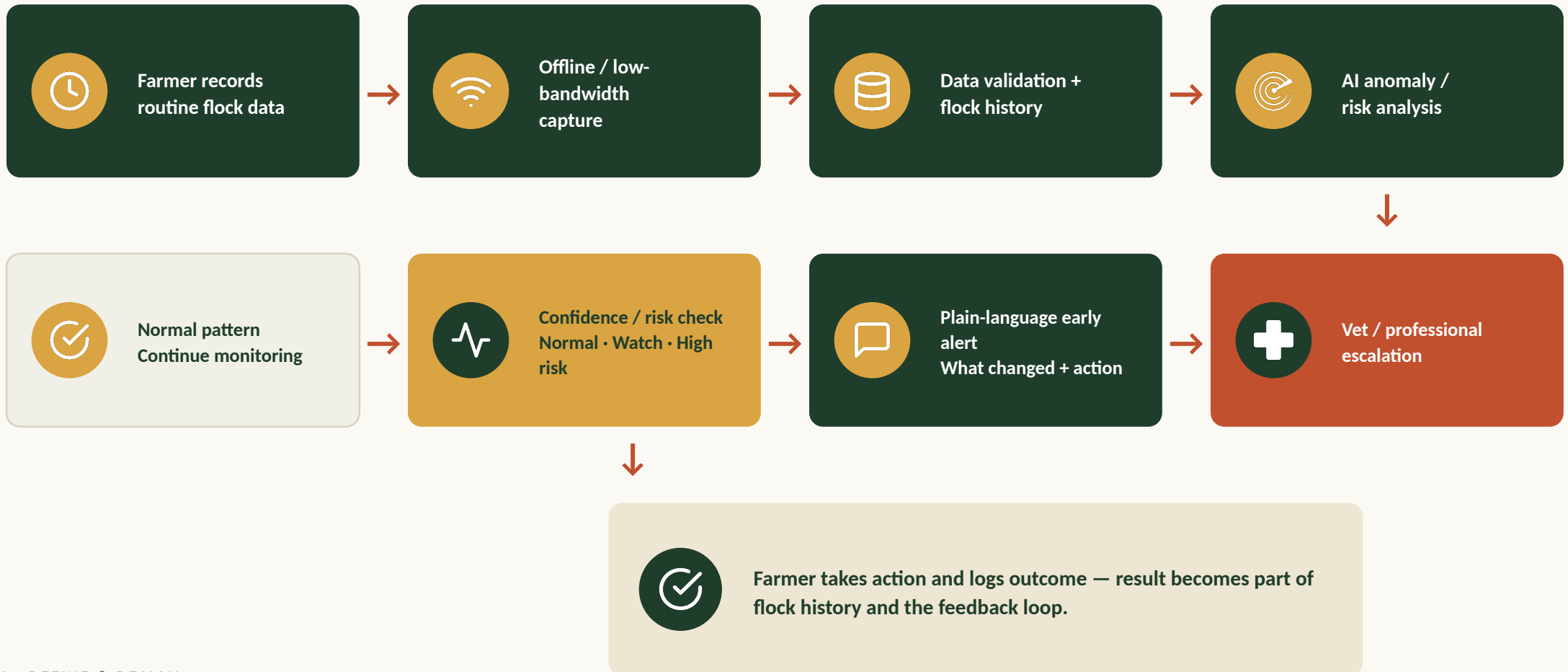
What changed

Recommended action

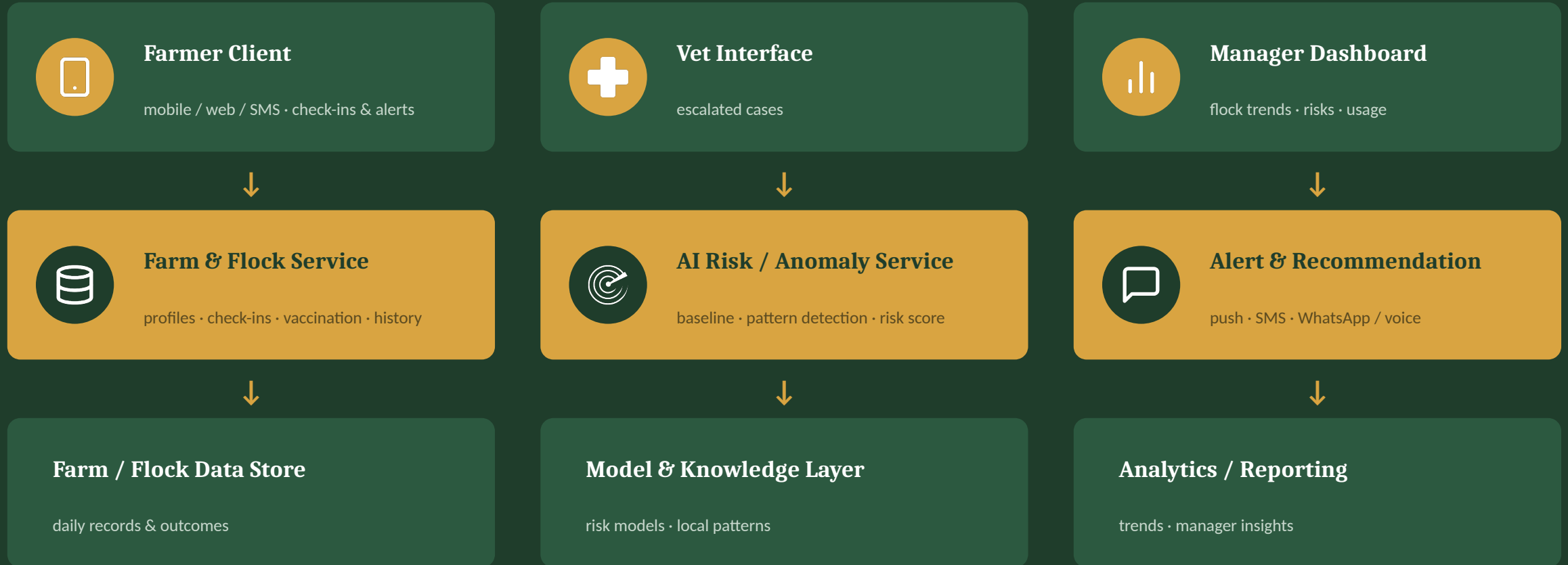
Contact a vet

Continuous flock monitoring & early warning

Updated for continuous flock monitoring and early warning.



Three layers: client, services, data



Architecture flow: Farmer data → sync / API → farm services + AI analysis → alerts / recommendations → outcome logging → stored history & analytics.

Functional & non-functional requirements



Functional

- Register/login and create farms/flocks
- Record daily or periodic flock data
- View flock health & production trends
- Receive early-warning alerts & actions
- Log actions and outcomes after an alert
- Receive vaccination / check-in reminders
- Escalate cases to a health professional
- Manager view across multiple flocks



Non-Functional

- Affordable for small/medium-scale farms
- Simple interface, minimal data-entry burden
- Offline-friendly / unstable-connectivity tolerant
- Secure handling of farm and user data
- Fast alert delivery for high-risk patterns
- Explainable alerts: what changed & why
- AI supports triage — not a guaranteed diagnosis

Open questions for Week 2

The solution design is still a working hypothesis. Week 2 should test whether farmers will provide routine data, trust the alerts, and act on them quickly enough to create value.

1

Do farmers currently record mortality, feed, water, egg output or growth? If yes, how?

3

How often would they realistically check in — daily, every few days, or weekly?

5

Would they trust an AI alert if it explains the reason behind the warning?

7

How much did their most recent disease/mortality event cost?

9

When should the system say "contact a vet" rather than self-monitor?

2

Which 2-4 data points are easiest for them to record consistently?

4

What warning would make them take immediate action?

6

Do they prefer mobile app, web app, WhatsApp, SMS, or a combination?

8

Would they pay monthly for a tool that demonstrably reduces losses? At what price?

WEEK 2 MILESTONE

Demonstrate that AI is the right solution.

The strongest case for AI is not simply identifying a disease from a photo. The vertical SaaS becomes more valuable when AI continuously analyzes flock data, detects patterns difficult to notice manually, prioritizes risk, and helps the farmer decide what to do next.

BEFORE MOVING TO WEEK 3, VALIDATE:

-  The problem is frequent and financially significant for target farmers.
-  Farmers can and will provide enough routine data for useful monitoring.
-  AI adds meaningful value over simple reminders or static rules.
-  Farmers understand and trust the alert format.
-  The MVP can be built affordably and used under realistic connectivity.